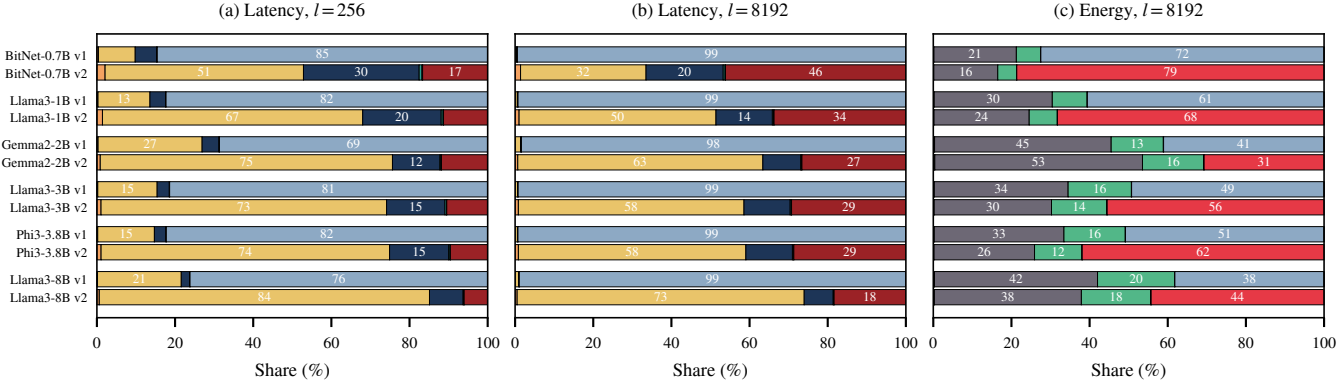
(a) Latency, $l = 256$
(b) Latency, $l = 8192$
(c) Energy, $l = 8192$
BitNet-0.7B v1
BitNet-0.7B v2
Llama3-1B v1
Llama3-1B v2
Gemma2-2B v1
Gemma2-2B v2
Llama3-3B v1
Llama3-3B v2
Phi3-3.8B v1
Phi3-3.8B v2
Llama3-8B v1
Llama3-8B v2
0
20
40
60
80
100
Share (%)
Latency (a, b):
Xbar+DAC+ADC
Communication
Buffer
Digital
Attn: systolic array (v1)
Attn: PIM xbar (v2)
Attn: PIM buffer (v2)
Energy (c):
Crossbar
DAC
ADC
Buffer
Other digital
Attn: systolic array (v1)
Attn: PIM (v2)